CS23532 — COMPUTER NETWORKS LAB

PRACTICAL 2: TYPES OF NETWORK CABLES

Aim: To learn about the different cables used in computer networks and understand the basic procedure for preparing Ethernet straight-through and crossover cables.

a) Study of Network Cable Types

Networking uses several kinds of cables to carry data between devices. The main types considered in this experiment are Unshielded Twisted Pair (UTP), Shielded Twisted Pair (STP), Screened Shielded Twisted Pair (SSTP), coaxial cable, and fibre-optic cable.

Cable type	Category / type	Maximum data rate (as listed)	Advantages and limitations	Applications
UTP	Cat 3	10 Mbps	Low cost and simple to install because of its small diameter. More affected by electromagnetic interference and noise.	10Base-T Ethernet
UTP	Cat 5	Up to 100 Mbps	Economical and easy to handle; vulnerable to interference.	Fast Ethernet; Gigabit Ethernet (as noted in source)
UTP	Cat 5e	1 Gbps	Convenient to install; shielding is not provided, so interference can be a concern.	Fast Ethernet; Gigabit Ethernet
STP	Cat 6 / 6a	10 Gbps	Shielding improves resistance to noise and interference and supports higher speeds than UTP. It costs more and takes greater effort to install.	Gigabit Ethernet; 10G Ethernet (55 m in the source table)
SSTP	Cat 7	10 Gbps	Uses additional shielding to reduce interference; generally more costly and harder to install.	10G Ethernet (100 m is noted in the source table)
Coaxial	RG-6, RG-59, RG-11	10–100 Mbps	Offers high bandwidth, low signal loss and resistance to interference. Limitations include bulk, cost and distance constraints.	Television networks and high-speed internet connections
Fibre optic	Single-mode / multi-mode	100 Gbps	High speed, large bandwidth, strong security and long-distance transmission. It is expensive and requires skilled installation.	High-speed and long-distance communication

b) Making an Ethernet Straight-Through or Crossover Cable

Tools and materials: CAT5 or CAT5e Ethernet cable, a crimping tool, two RJ45 connectors, and optional connector boots/shields. CAT5e is suitable for Gigabit Ethernet, while CAT5 may be used over shorter distances.

Difference between the cable types: A straight-through cable uses the same wiring standard at both ends (A–A or B–B). A crossover cable uses one wiring standard at one end and the other standard at the opposite end (A–B). Follow the wiring diagram provided by the lab manual when arranging the conductors.

Procedure

Step 1. Slide the optional connector boots onto the cable before attaching the RJ45 plugs.

Step 2. Use the crimping tool to remove about 1.5 cm of the outer cable jacket at each end, taking care not to damage the internal wires.

Step 3. Separate and straighten the four twisted pairs. Arrange the eight wires in the required order using the T568A or T568B reference diagram.

Step 4. Bring the wires into a flat row and trim their ends evenly so that they can enter the RJ45 connector together.

Step 5. Hold the RJ45 plug with the contacts facing toward you and the clip facing away. Insert the wires carefully, maintaining the correct sequence, and ensure the outer jacket reaches the plug's retaining notch.

Step 6. If the jacket does not reach the notch, trim and prepare the cable end again as needed. Check that each conductor reaches the front of the connector.

Step 7. Place the connector into the crimping tool and press firmly to secure the pins. Avoid excessive force, which may damage the connector.

Step 8. Repeat the process at the other end. Use the same wiring standard at both ends for a straight-through cable, or use T568A at one end and T568B at the other for a crossover cable.

Step 9. Connect the cable to the intended devices and test the connection.

Student Observation / Questions

1. What is the main difference between a crossover cable and a straight-through cable?

2. Which cable type (straight-through or crossover) is used to connect two PCs directly in the setup being studied?

3. Which cable type is used to connect a router or switch to a PC in the setup being studied?

4. Identify the category of twisted-pair cable used in your laboratory to connect a PC to the network socket.

5. Describe what you learned, the difficulties encountered while making the cable, and the result obtained after testing it.

Result

The different types of network cables and their features, advantages, limitations, and uses were studied. The procedure for arranging wires and preparing Ethernet straight-through and crossover cables using RJ45 connectors was also reviewed.

Conclusion

This practical introduced common network transmission media and the basic steps involved in making Ethernet cables. Correct wire ordering, careful crimping, and testing are important to ensure a reliable cable connection.